Github Link: https://github.com/abargeer/CS5782-FinalProject

Introduction & Chosen Result

Early neural captioning models compressed an entire image into a single vector via a CNN before decoding with an RNN, creating an information bottleneck that produced generic captions. Xu et al. [1] addressed this in "Show, Attend and Tell" (ICML 2015) by introducing attention-based encoder-decoder captioning. A pretrained CNN extracts a spatial feature map (14 x 14 grid), and at each decoding step an attention mechanism computes weights over spatial locations to produce a context vector, allowing an LSTM decoder to selectively focus on relevant regions. The paper proposes soft (deterministic) attention trained via backpropagation with doubly stochastic regularization, and hard (stochastic) attention trained via REINFORCE. Both achieve strong results on Flickr8k[2], Flickr30k[3], and MS COCO[4], with attention weights yielding interpretable visualizations. We reimplement the soft attention model in PyTorch[5] and extend the paper with two experimental probes testing whether spatially grounded attention is essential for caption quality.

We target the soft attention results on Flickr30k from Table 1 of the paper. This captures the paper's primary contribution of showing that attention over spatial CNN features improves both caption quality and interpretability. Flickr30k is the paper's main benchmark (31K images, 5 captions each) and is trainable on Colab GPUs.

Methodology

Following the paper, our encoder uses a frozen, pretrained VGG-19 [3] to extract 14×14× 512 spatial features from a deep convolutional layer, yielding 196 annotation vectors. The attention module computes additive alignment scores between each annotation and the previous LSTM hidden state, normalizes via softmax, and produces a context vector fed into an LSTM decoder. The decoder is a single-layer LSTM (hidden dim 512) with a deep output layer. Full architectural details match Section 3.1 and Equations 1-7 of the paper. We use Adam [5] (lr = 4×10^{-4}), batch size 64, gradient clipping at 5.0, and dropout 0.5. The loss is cross-entropy plus the doubly stochastic regularization penalty from Eq. 14 ($\lambda = 1.0$). Vocabulary size is 10,000 words. We apply early stopping on validation BLEU-4 and use beam search (k = 3) at inference. Metrics are computed via pycocoevalcap [7]: BLEU-1 through 4 and METEOR. Images are resized (shortest side 256), center-cropped to 224 × 224, with random cropping and horizontal flipping during training. Data is split 85/10/5 for train/val/test.

To investigate whether attention's benefit comes from spatial grounding or feature mixing, we trained two controlled variants sharing the same architecture and hyperparameters, isolating attention as the only variable. The first is Shuffled Attention: at each timestep (train and inference), attention weights are randomly permuted across spatial locations before computing the context vector. The second is Entropy Regularized/Sharpened Attention: an additional penalty encourages sharper attention, applied selectively on timesteps generating object nouns. This tests whether forcing stronger spatial commitment improves content-word accuracy.

Results & Analysis

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4	METEOR

Paper (Soft Attention)	66.7	43.4	28.8	19.1	18.49
Our Baseline (Soft)	62.1	41.9	29.0	20.1	20.85
Shuffled Attention	58.1	37.3	24.6	16.5	19.05
Entropy Regularized Attention	58.3	38.4	25.8	17.5	19.09

Table 1: Flickr30 results compared to the paper's reported soft attention scores

Our baseline reimplementation closely matches the paper. We exceed it on BLEU-4 and METEOR, while falling short on BLEU-1. The BLEU-1 gap likely stems from preprocessing differences or vocabulary construction. The strong METEOR suggests semantically appropriate captions despite lower unigram overlap. For Shuffled Attention, destroying spatial correspondence caused consistent drops in BLEU scores, confirming that spatial grounding is genuinely beneficial. However, the shuffled model still produces reasonable captions, indicating attention also provides useful feature mixing independent of spatial alignment. For Entropy-Regularized (Sharp) Attention, sharpening attention on object nouns did not improve results. The soft attention mechanism already learns appropriate focus through doubly stochastic regularization. Forcing additional sharpness over-constrains the model, preventing it from leveraging contextual information from surrounding regions. We observed that failure cases commonly stem from the decoder relying on common training co-occurrences rather than attention looking at the wrong regions. The model also struggles with images very different from the training data (e.g., abstract scenes).

Figure 1. Caption: "A man in a red jacket is sitting on a bench". Attention visualizations show the baseline model correctly attends to objects for content words ("man", "jacket", "red") but spreads diffusely for function words ("a", "the", "in") consistent with the original paper.

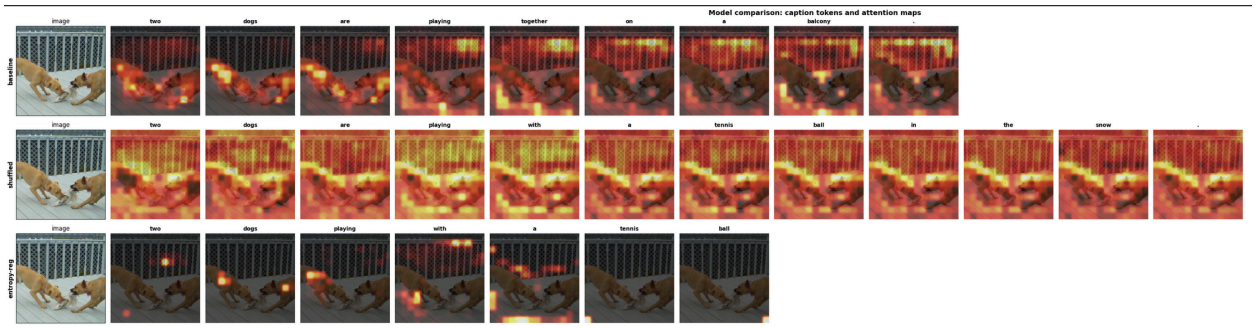

Figure 2. Top (Baseline): “Two dogs are playing together on a balcony.” Middle (Shuffled Attention): “Two dogs are playing with a tennis ball in the snow.” Bottom (Entropy-Regularized/Sharp Attention): “Two dogs playing with a tennis ball”. The other two models hallucinate, while the baseline caption most accurately describes the image. The Sharpened Attention focuses much more on specific parts of the image, while the Shuffled Attention focuses everywhere.

Reflections

From the shuffled experiment, we discovered that attention genuinely benefits from spatial correspondence. From the entropy-regularized experiment, we discovered that more focus is not always better, as the model benefits from integrating context across regions. Overall, our key takeaway is that most errors originate from language model biases rather than attention failures, suggesting decoder improvements or more training data would yield larger gains than refining attention. One major challenge we encountered was managing Colab session limits during Flickr30k training. This required checkpointing after each epoch and pre-extracting VGG-19 features to avoid redundant computation. Additionally, matching BLEU-1 precisely proved difficult without the authors’ exact preprocessing and hyperparameter search. Future work includes implementing hard attention via REINFORCE, using modern encoders (ResNet, ViT) to assess whether stronger features reduce decoder reliance on language priors, and scaling to MS COCO (82K images).

References

[1] Xu, K., Ba, J.L., Kiros, R., Cho, K., Courville, A., Salakhutdinov, R., Zemel, R.S., & Bengio, Y. (2015). Show, Attend and Tell: Neural Image Caption Generation with Visual Attention. ICML.
[2] Flickr8k: https://www.kaggle.com/datasets/adityajn105/flickr8k
[3] Flickr30k: https://www.kaggle.com/datasets/adityajn105/flickr30k
[4] Chen, X. et al. (2015). Microsoft COCO Captions: Data Collection and Evaluation Server. arXiv:1504.00325.
[5] Paszke, A. et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. NeurIPS.
[6] Simonyan, K. & Zisserman, A. (2015). Very Deep Convolutional Networks for Large-Scale Image Recognition. ICLR.
[7] Kingma, D.P. & Ba, J. (2015). Adam: A Method for Stochastic Optimization. ICLR.